7 Oefeningen

1 Los de volgende vergelijkingen op in $\mathbb{Q}$.

a
$$17+x = 34 \Updownarrow x = 34-17 \Updownarrow x = 17$$

b
$$x-0,5 = 7,2 \Updownarrow x = 7,2+0,5 \Updownarrow x = 7,7$$

c
$$y+\frac{3}{5} = \frac{-2}{5} \Updownarrow y = -\frac{2}{5}-\frac{3}{5} \Updownarrow y = -\frac{5}{5} \Updownarrow y = -1$$

d
$$k-\frac{1}{4} = 1,25 \Updownarrow k = 1,25+0,25 \Updownarrow k = 1,5$$

e
$$2+x = \frac{-1}{3} \Updownarrow x = -\frac{1}{3}-2 \Updownarrow x = -\frac{1}{3}-\frac{6}{3} \Updownarrow x = -\frac{7}{3}$$

f
$$\frac{4}{3}+x = \frac{8}{3} \Updownarrow x = \frac{8}{3}-\frac{4}{3} \Updownarrow x = \frac{4}{3}$$

g
$$t-\frac{11}{4} = -\frac{3}{2} \Updownarrow t = -\frac{3}{2}+\frac{11}{4} \Updownarrow t = -\frac{6}{4}+\frac{11}{4} \Updownarrow t = \frac{5}{4}$$

h
$$0,24+x = -5,17 \Updownarrow x = -5,17-0,24 \Updownarrow x = -5,41$$

i
$$y-\frac{1}{8} = \frac{5}{4} \Updownarrow y = \frac{5}{4}+\frac{1}{8} \Updownarrow y = \frac{10}{8}+\frac{1}{8} \Updownarrow y = \frac{11}{8}$$

2 Los de volgende vergelijkingen op in $\mathbb{Q}$.

a
$$-4x = 44 \Updownarrow x = 44:(-4) \Updownarrow x = -11$$

b
$$2y = -0,36 \Updownarrow y = -0,36:2 \Updownarrow y = -0,18$$

c
$$\frac{1}{3}x = \frac{5}{2} \Updownarrow x = \frac{5}{2}:\frac{1}{3} \Updownarrow x = \frac{5}{2}\cdot\frac{3}{1} \Updownarrow x = \frac{15}{2}$$

d
$$\frac{3}{14}x = \frac{-1}{7} \Updownarrow x = -\frac{1}{7}:\frac{3}{14} \Updownarrow x = -\frac{1}{7}\cdot\frac{14}{3} \Updownarrow x = -\frac{2}{3}$$

e
$$0,25x = 0,5 \Updownarrow x = 0,5:0,25 \Updownarrow x = 2$$

f
$$-11x = -\frac{11}{13} \Updownarrow x = -\frac{11}{13}:(-11) \Updownarrow x = -\frac{11}{13}\cdot\frac{-1}{11} \Updownarrow x = \frac{1}{13}$$

3 Los de volgende vergelijkingen op in $\mathbb{Q}$.

a
$$x-3,75 = -2,4$$
$$\Updownarrow$$
$$x = -2,4+3,75$$
$$\Updownarrow$$
$$x = 1,35$$

b
$$\frac{1}{3}x = 9$$
$$\Updownarrow$$
$$x = 9\cdot 3$$
$$\Updownarrow$$
$$x = 27$$

c
$$4x = -\frac{17}{2}$$
$$\Updownarrow$$
$$x = -\frac{17}{2}:4$$
$$\Updownarrow$$
$$x = -\frac{17}{2}\cdot\frac{1}{4}$$
$$\Updownarrow$$
$$x = -\frac{17}{8}$$

d
$$x+\frac{3}{5} = 4$$
$$\Updownarrow$$
$$x = 4-\frac{3}{5}$$
$$\Updownarrow$$
$$x = \frac{20}{5}-\frac{3}{5}$$
$$\Updownarrow$$
$$x = \frac{17}{5}$$

e
$$x-\frac{7}{2} = 3,5$$
$$\Updownarrow$$
$$x = 3,5+3,5$$
$$\Updownarrow$$
$$x = 7$$

f
$$\frac{5}{7}k = -\frac{1}{14}$$
$$\Updownarrow$$
$$x = -\frac{1}{14}:\frac{5}{7}$$
$$\Updownarrow$$
$$x = -\frac{1}{14}\cdot\frac{7}{5}$$
$$\Updownarrow$$
$$x = -\frac{1}{10}$$

g
$$\frac{1}{12}+k = -\frac{2}{9}$$
$$\Updownarrow$$
$$k = -\frac{2}{9}-\frac{1}{12}$$
$$\Updownarrow$$
$$k = -\frac{8}{36}-\frac{3}{36}$$
$$\Updownarrow$$
$$k = -\frac{11}{36}$$

h
$$0,25+a = \frac{7}{8}$$
$$\Updownarrow$$
$$a = \frac{7}{8}-\frac{1}{4}$$
$$\Updownarrow$$
$$a = \frac{7}{8}-\frac{2}{8}$$
$$\Updownarrow$$
$$a = \frac{5}{8}$$

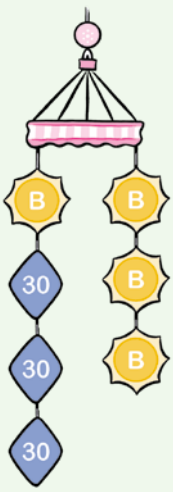
i
$$-\frac{5}{12}x = \frac{10}{3}$$
$$\Updownarrow$$
$$x = \frac{10}{3}:\left(-\frac{5}{12}\right)$$
$$\Updownarrow$$
$$x = \frac{10}{3}\cdot\left(-\frac{12}{5}\right)$$
$$\Updownarrow$$
$$x = -8$$

4 In de babykamer van Adil hangen speeltjes die perfect in evenwicht zijn. Op de stukjes staat het gewicht in gram. Bepaal het gewicht van het stukje met een letter op.

a

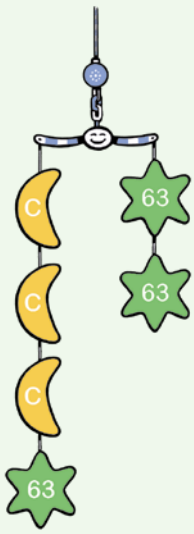
$$2a = 40$$
$$\Updownarrow$$
$$a = 20$$

b



$$90 = 2b$$
$$\Updownarrow$$
$$45 = b$$

c



$$3c = 63$$
$$\Updownarrow$$
$$c = 21$$

5 Los de volgende vergelijkingen op in $\mathbb{Q}$.

a
$$-x-3 = 2+2x$$
$$\Updownarrow$$
$$-3-2 = 2x+x$$
$$\Updownarrow$$
$$-5 = 3x$$
$$\Updownarrow$$
$$-\frac{5}{3} = x$$

b
$$-4+4x = -4+x$$
$$\Updownarrow$$
$$4x-x = -4+4$$
$$\Updownarrow$$
$$3x = 0$$
$$\Updownarrow$$
$$x = 0$$

c
$$20x-10 = 28+x$$
$$\Updownarrow$$
$$20x-x = 28+10$$
$$\Updownarrow$$
$$19x = 38$$
$$\Updownarrow$$
$$x = \frac{38}{19}$$
$$\Updownarrow$$
$$x = 2$$

d
$$2+x = 4-3x$$
$$\Updownarrow$$
$$x+3x = 4-2$$
$$\Updownarrow$$
$$4x = 2$$
$$\Updownarrow$$
$$x = \frac{2}{4}$$
$$\Updownarrow$$
$$x = \frac{1}{2}$$

e
$$2x-48 = -13x+12$$
$$\Updownarrow$$
$$2x+13x = 12+48$$
$$\Updownarrow$$
$$15x = 60$$
$$\Updownarrow$$
$$x = 4$$

f
$$2x-8 = x+2$$
$$\Updownarrow$$
$$2x-x = 2+8$$
$$\Updownarrow$$
$$x = 10$$

g
$$-3x+8 = 5x$$
$$\Updownarrow$$
$$8 = 5x+3x$$
$$\Updownarrow$$
$$8 = 8x$$
$$\Updownarrow$$
$$1 = x$$

h
$$x-15 = -3x+1$$
$$\Updownarrow$$
$$x+3x = 1+15$$
$$\Updownarrow$$
$$4x = 16$$
$$\Updownarrow$$
$$x = 4$$

6 Los de volgende vergelijkingen op in $\mathbb{Q}$.

a
$$\frac{3}{2}x+4 = \frac{1}{2}x-2$$
$$\Updownarrow$$

$$\begin{aligned} \frac{3}{2}x-\frac{1}{2}x &= -4-2 \\ &\Updownarrow \\ x &= -6 \end{aligned}$$

$$\begin{aligned} 3x+8 &= x-4 \\ &\Updownarrow \\ 3x-x &= -4-8 \\ &\Updownarrow \\ 2x &= -12 \\ &\Updownarrow \\ x &= -6 \end{aligned}$$

b
$$\frac{1}{7}x+\frac{1}{2} = \frac{1}{2}x-2$$
$$\Updownarrow$$

$$\begin{aligned} \frac{1}{7}x-\frac{1}{2}x &= -2-\frac{1}{2} \\ &\Updownarrow \\ 2x-7x &= -28-7 \\ &\Updownarrow \\ -5x &= -35 \\ &\Updownarrow \\ x &= 7 \end{aligned}$$

$$\begin{aligned} \frac{1}{7}x-\frac{1}{2}x &= -2-\frac{1}{2} \\ &\Updownarrow \\ \frac{2}{14}x-\frac{7}{14}x &= -\frac{4}{2}-\frac{1}{2} \\ &\Updownarrow \\ -\frac{5}{14}x &= -\frac{5}{2} \\ &\Updownarrow \\ x &= -\frac{5}{2}:\left(-\frac{5}{14}\right) \\ &\Updownarrow \\ x &= -\frac{5}{2}\cdot\left(-\frac{14}{5}\right) \\ &\Updownarrow \\ x &= 7 \end{aligned}$$

c
$$0,4x-0,6x = 0,8$$
$$\Updownarrow$$

$$\begin{aligned} 4x-6x &= 8 \\ &\Updownarrow \\ -2x &= 8 \\ &\Updownarrow \\ x &= -4 \end{aligned}$$

$$\begin{aligned} -0,2x &= 0,8 \\ &\Updownarrow \\ x &= 0,8:(-0,2) \\ &\Updownarrow \\ x &= -4 \end{aligned}$$

d
$$\frac{3}{2}x-\frac{1}{6} = \frac{4}{3}$$
$$\Updownarrow$$

$$\begin{aligned} 9x-1 &= 8 \\ &\Updownarrow \\ 9x &= 9 \\ &\Updownarrow \\ x &= 1 \end{aligned}$$

$$\begin{aligned} \frac{3}{2}x &= \frac{4}{3}+\frac{1}{6} \\ &\Updownarrow \\ \frac{3}{2}x &= \frac{8}{6}+\frac{1}{6} \\ &\Updownarrow \\ \frac{3}{2}x &= \frac{3}{2} \\ &\Updownarrow \\ x &= 1 \end{aligned}$$

e
$$x-\frac{3}{2} = \frac{1}{2}x+\frac{2}{3}$$
$$\Updownarrow$$

$$\begin{aligned} 6x-9 &= 3x+4 \\ &\Updownarrow \\ 6x-3x &= 9+4 \\ &\Updownarrow \\ 3x &= 13 \\ &\Updownarrow \\ x &= \frac{13}{3} \end{aligned}$$

$$\begin{aligned} x-\frac{1}{2}x &= \frac{2}{3}+\frac{3}{2} \\ &\Updownarrow \\ \frac{2}{2}x-\frac{1}{2}x &= \frac{4}{6}+\frac{9}{6} \\ &\Updownarrow \\ \frac{1}{2}x &= \frac{13}{6} \\ &\Updownarrow \\ x &= \frac{13}{6}\cdot 2 \\ &\Updownarrow \\ x &= \frac{13}{3} \end{aligned}$$

f
$$\frac{2}{3}x+4 = \frac{1}{2}x-2$$
$$\Updownarrow$$

$$\begin{aligned} 4x+24 &= 3x-12 \\ &\Updownarrow \\ 4x-3x &= -12-24 \\ &\Updownarrow \\ x &= -36 \end{aligned}$$

$$\begin{aligned} \frac{2}{3}x-\frac{1}{2}x &= -2-4 \\ &\Updownarrow \\ \frac{4}{6}x-\frac{3}{6}x &= -2-4 \\ &\Updownarrow \\ \frac{1}{6}x &= -6 \\ &\Updownarrow \\ x &= -36 \end{aligned}$$

g
$$\frac{-9}{2}x-\frac{4}{3} = \frac{1}{2}x+\frac{7}{2}$$
$$\Updownarrow$$

$$\begin{aligned} -27x-8 &= 3x+21 \\ &\Updownarrow \\ -27x-3x &= 21+8 \\ &\Updownarrow \\ -30x &= 29 \\ &\Updownarrow \\ x &= -\frac{29}{30} \end{aligned}$$

$$\begin{aligned} -\frac{9}{2}x-\frac{1}{2}x &= \frac{7}{2}+\frac{4}{3} \\ &\Updownarrow \\ -\frac{10}{2}x &= \frac{21}{6}+\frac{8}{6} \\ &\Updownarrow \\ -\frac{10}{2}x &= \frac{29}{6} \\ &\Updownarrow \\ x &= \frac{29}{6}\cdot\left(-\frac{2}{10}\right) \\ &\Updownarrow \\ x &= -\frac{29}{30} \end{aligned}$$

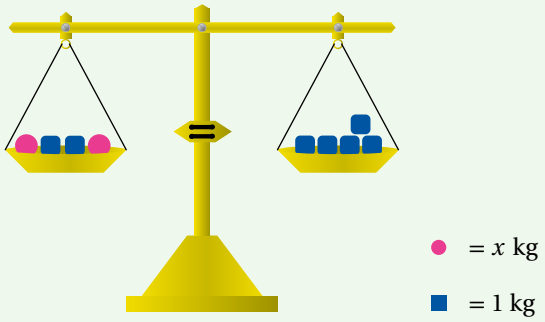
h
$$0,55x+0,18 = 0,35x-0,12$$
$$\Updownarrow$$

$$\begin{aligned} 55x+18 &= 35x-12 \\ &\Updownarrow \\ 55x-35x &= -12-18 \\ &\Updownarrow \\ 20x &= -30 \\ &\Updownarrow \\ x &= -\frac{30}{20} \\ &\Updownarrow \\ x &= -\frac{3}{2} \end{aligned}$$

$$\begin{aligned} 0,55x-0,35x &= -0,12-0,18 \\ &\Updownarrow \\ 0,2x &= -0,3 \\ &\Updownarrow \\ x &= -0,3:0,2 \\ &\Updownarrow \\ x &= -\frac{3}{2} \end{aligned}$$

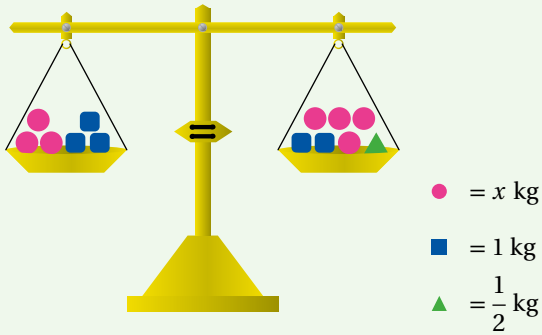
7 Stel telkens de vergelijking op en los ze nadien op.

a



$$\begin{aligned} 2x+2 &= 5 \\ &\Updownarrow \\ 2x &= 3 \\ &\Updownarrow \\ x &= \frac{3}{2} \end{aligned}$$

b



$$\begin{aligned} 3x+3 &= 4x+2{,}5 \\ &\Updownarrow \\ 3-2{,}5 &= 4x-3x \\ &\Updownarrow \\ 0{,}5 &= x \end{aligned}$$

8 Zoek x met behulp van volgende pijlvoorstelling.

a

$x \xrightarrow{\cdot 2} \ldots 10 \ldots \xrightarrow{-3} 7$

$x = 5$

$$\begin{aligned} 2x-3 &= 7 \\ &\Updownarrow \\ 2x &= 7+3 \\ &\Updownarrow \\ 2x &= 10 \\ &\Updownarrow \\ x &= 5 \end{aligned}$$

b

$x \xrightarrow{\cdot(-4)} \ldots -4 \ldots \xrightarrow{+11} 7$

$x = 1$

$$\begin{aligned} -4x+11 &= 7 \\ &\Updownarrow \\ 11-7 &= 4x \\ &\Updownarrow \\ 4 &= 4x \\ &\Updownarrow \\ 1 &= x \end{aligned}$$

c

$x \xrightarrow{\cdot\frac{1}{2}} \ldots \frac{1}{2} \ldots \xrightarrow{+\frac{9}{4}} \frac{11}{4}$

$x = 1$

$$\begin{aligned} \frac{1}{2}\cdot x+\frac{9}{4} &= \frac{11}{4} \\ &\Updownarrow \\ \frac{1}{2}x &= \frac{1}{2} \\ &\Updownarrow \\ x &= 1 \end{aligned}$$

d

$x \xrightarrow{\cdot\left(-\frac{2}{3}\right)} \ldots -2 \ldots \xrightarrow{-\frac{5}{2}} \frac{-9}{2}$

$x = 3$

$$\begin{aligned} \left(-\frac{2}{3}\right)\cdot x-\frac{5}{2} &= -\frac{9}{2} \\ &\Updownarrow \\ -\frac{5}{2}+\frac{9}{2} &= \frac{2}{3}x \\ &\Updownarrow \\ 2 &= \frac{2}{3}x \\ &\Updownarrow \\ 2:\frac{2}{3} &= x \\ &\Updownarrow \\ 3 &= x \end{aligned}$$

9 Los de volgende vergelijkingen op in $\mathbb{Q}$.

a
$$7\cdot(p-2) = 14$$
$$\Updownarrow$$
$$\begin{aligned} 7p-14 &= 14 \\ &\Updownarrow \\ 7p &= 28 \\ &\Updownarrow \\ p &= 4 \end{aligned}$$

b
$$18 = 3-(-x-5)$$
$$\Updownarrow$$
$$\begin{aligned} 18 &= 3+x+5 \\ &\Updownarrow \\ 18-3-5 &= x \\ &\Updownarrow \\ 10 &= x \end{aligned}$$

c
$$3\cdot(2k-1) = 5\cdot(k-4)$$
$$\Updownarrow$$
$$\begin{aligned} 6k-3 &= 5k-20 \\ &\Updownarrow \\ k &= -17 \end{aligned}$$

d
$$(4x-2)\cdot 5 = 27$$
$$\Updownarrow$$
$$\begin{aligned} 20x-10 &= 27 \\ &\Updownarrow \\ 20x &= 37 \\ &\Updownarrow \\ x &= \frac{37}{20} \end{aligned}$$

e
$$-4\cdot(3-x)+5x = 15$$
$$\Updownarrow$$
$$\begin{aligned} -12+4x+5x &= 15 \\ &\Updownarrow \\ 9x &= 15+12 \\ &\Updownarrow \\ 9x &= 27 \\ &\Updownarrow \\ x &= 3 \end{aligned}$$

f
$$-3\cdot(m+0,5) = -5\cdot(m-0,3)$$
$$\Updownarrow$$
$$\begin{aligned} -3m-1,5 &= -5m+1,5 \\ &\Updownarrow \\ -3m+5m &= 1,5+1,5 \\ &\Updownarrow \\ 2m &= 3 \\ &\Updownarrow \\ m &= \frac{3}{2} \end{aligned}$$

$$2x = -12 \Leftrightarrow x = -6$$

10 Welke vergelijking heeft dezelfde oplossing als $2x + 3 = -9$?

A	B	C	D
$-4x - 4 = -2x + 2$	$3x - 5 = 13$	$\frac{1}{2}x + \frac{9}{5} = \frac{1}{5}x$	$\frac{3}{2}x - \frac{1}{2} = \frac{5}{2}x + 7$
$\Updownarrow$	$\Updownarrow$	$\Updownarrow$	$\Updownarrow$
$-4 - 2 = -2x + 4x$	$3x = 13 + 5$	$5x + 18 = 2x$	$3x - 1 = 5x + 14$
$\Updownarrow$	$\Updownarrow$	$\Updownarrow$	$\Updownarrow$
$-6 = 2x$	$3x = 18$	$3x = -18$	$-15 = 2x$
$\Updownarrow$	$\Updownarrow$	$\Updownarrow$	$\Updownarrow$
$-3 = x$	$x = 6$	$x = -6$	$-7{,}5 = x$

11 Welke vergelijkingen hebben als oplossing 3?
Werk uit en zet de vergelijkingen die 3 als oplossing hebben in een fluokleurtje. Bij elke vergelijking staat een woord. Met de fluowoorden maak je een mooi spreekwoord.

$2x + 4 = 10$ KLEINE	$3 \cdot (x - 2) = 4 - (x - 2)$ NIET	$\frac{1}{3}x + 5 = 2x$ LEERT
$4x + 3x - (x + 1) = 8$ ~~KLOKJE~~	$-5 \cdot (x + 2) = -10 - (x + 12)$ HET	$3x - 3 = 0$ ~~THUIS~~
$6x + 4 - 3 \cdot (x + 2) = 7$ GROTE	$10x - 10 = 20$ HET	$\frac{2}{5}x + \frac{1}{2} = \frac{3}{5}x - \frac{1}{3}$ ~~NERGENS~~
$-2{,}5 \cdot x = -7{,}5$ DOET	$\frac{x}{2} + \frac{x}{3} = \frac{5}{2}$ WIE	$-2 \cdot (x + 1) = -6 + (x - 5)$ VERKEERD

Met de gevonden woorden kan ik dit spreekwoord maken:

Wie het kleine niet leert , doet het grote verkeerd.

12 Los de volgende vergelijkingen op in $\mathbb{Q}$.

a
$$-(x+6)+2\cdot(5x-3) = -13$$
$$\Updownarrow$$
$$\begin{aligned} -x-6+10x-6 &= -13 \\ &\Updownarrow \\ 9x-12 &= -13 \\ &\Updownarrow \\ 9x &= -1 \\ &\Updownarrow \\ x &= -\frac{1}{9} \end{aligned}$$

b
$$4\cdot(3b+2) = 6b+35$$
$$\Updownarrow$$
$$\begin{aligned} 12b+8 &= 6b+35 \\ &\Updownarrow \\ 12b-6b &= 35-8 \\ &\Updownarrow \\ 6b &= 27 \\ &\Updownarrow \\ b &= \frac{27}{6} \\ &\Updownarrow \\ b &= \frac{9}{2} \end{aligned}$$

c
$$14x-(9x-7) = 62$$
$$\Updownarrow$$
$$\begin{aligned} 14x-9x+7 &= 62 \\ &\Updownarrow \\ 5x &= 62-7 \\ &\Updownarrow \\ 5x &= 55 \\ &\Updownarrow \\ x &= 11 \end{aligned}$$

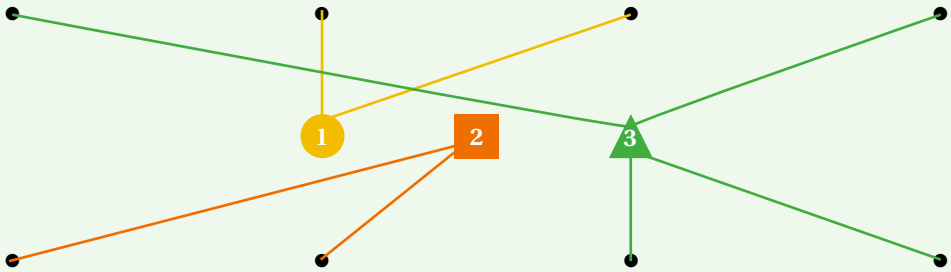
d
$$5\cdot(x+4)-3\cdot(x-2) = 32$$
$$\Updownarrow$$
$$\begin{aligned} 5x+20-3x+6 &= 32 \\ &\Updownarrow \\ 2x+26 &= 32 \\ &\Updownarrow \\ 2x &= 6 \\ &\Updownarrow \\ x &= 3 \end{aligned}$$

e
$$4\cdot(5y+3)+5\cdot(7-2y) = 0$$
$$\Updownarrow$$
$$\begin{aligned} 20y+12+35-10y &= 0 \\ &\Updownarrow \\ 10y &= -47 \\ &\Updownarrow \\ y &= -\frac{47}{10} \end{aligned}$$

f
$$7\cdot(x+2)-3\cdot(x-8)+2x = 3\cdot(4x+10)$$
$$\Updownarrow$$
$$\begin{aligned} 7x+14-3x+24+2x &= 12x+30 \\ &\Updownarrow \\ 6x+38 &= 12x+30 \\ &\Updownarrow \\ 6x-12x &= 30-38 \\ &\Updownarrow \\ -6x &= -8 \\ &\Updownarrow \\ x &= \frac{8}{6} \\ &\Updownarrow \\ x &= \frac{4}{3} \end{aligned}$$

13 Controleer met ICT of de getallen 1, 2 of 3 een oplossing zijn voor de vergelijkingen.
Verbind telkens de opgave met de oplossing.

$\frac{1}{3}x = 1$ $\quad$ $2 \cdot (x+1) + x = 5$ $\quad$ $6 - 2x = 4x$ $\quad$ $\frac{x+3}{3} = 4x - 10$



$4 + 3(x-1) = 7$ $\quad$ $5x - 8 = 2$ $\quad$ $\frac{x}{2} + \frac{5}{2} = x + 1$ $\quad$ $3x - 8 = 2x - 5$

14 Daan en Merel rijden, elk van bij hen thuis, met de fiets naar school. Ze wonen op dezelfde weg, maar Daan woont op 7 km van de school en Merel woont op 5,5 km van de school. Daan fietst met zo'n snelheid dat hij elke minuut 350 m aflegt. Merel is iets trager en rijdt met een snelheid van 200 m/min.
Om te achterhalen wanneer en op welke afstand van de school ze elkaar zullen treffen, moeten we volgende vergelijking oplossen (alle afstanden zijn omgezet in m).

$7000 - 350x = 5500 - 200x$

a Los de vergelijking op. Welke informatie heb je nu gekregen?

$$\begin{aligned} 7000 - 350x &= 5500 - 200x \\ &\Updownarrow \\ 7000 - 5500 &= -200x + 350x \\ &\Updownarrow \\ 1500 &= 150x \\ &\Updownarrow \\ 10 &= x \end{aligned}$$

ANTWOORD: Na 10 minuten ontmoeten ze elkaar.

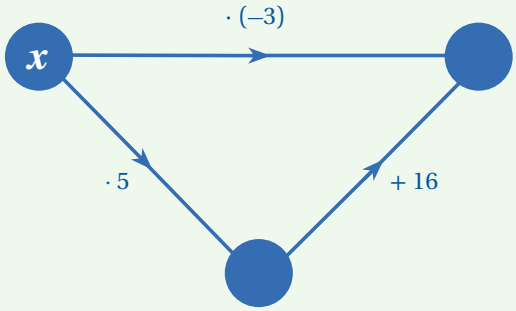
b Hoeveel meter zijn Daan en Merel op dat moment van school verwijderd?

$7000 - 350 \cdot 10 = 3500$ of ook: $5500 - 200 \cdot 10 = 3500$

ANTWOORD: Ze zijn dan 3,5 km van school verwijderd.

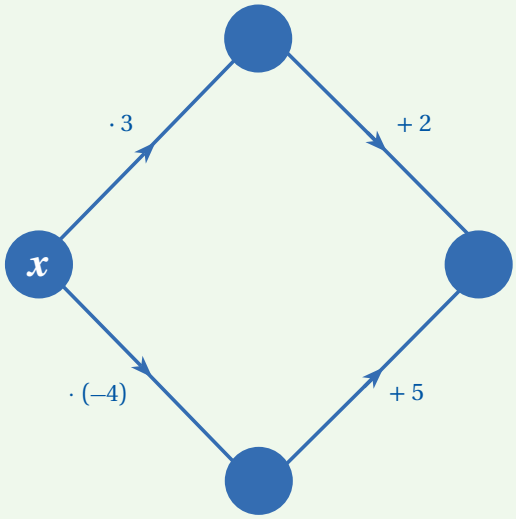
15 Bekijk onderstaande schema's. Welke vergelijking hoort hierbij? Los de vergelijking op.

a



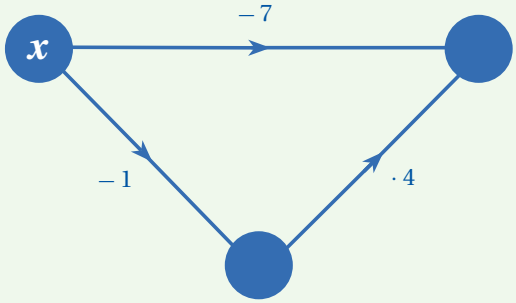
$$\begin{aligned} 5x+16 &= -3x \\ &\Updownarrow \\ 8x &= -16 \\ &\Updownarrow \\ x &= -2 \end{aligned}$$

b



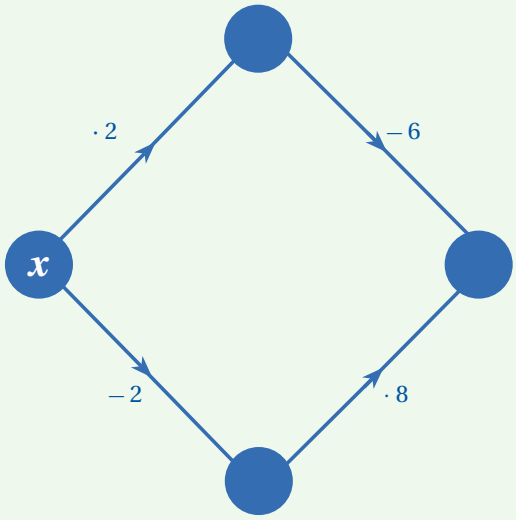
$$\begin{aligned} 3x+2 &= -4x+5 \\ &\Updownarrow \\ 7x &= 3 \\ &\Updownarrow \\ x &= \frac{3}{7} \end{aligned}$$

c



$$\begin{aligned} 4\cdot(x-1) &= x-7 \\ &\Updownarrow \\ 4x-4 &= x-7 \\ &\Updownarrow \\ 3x &= -3 \\ &\Updownarrow \\ x &= -1 \end{aligned}$$

d



$$\begin{aligned} 2x-6 &= (x-2)\cdot 8 \\ &\Updownarrow \\ 2x-6 &= 8x-16 \\ &\Updownarrow \\ 10 &= 6x \\ &\Updownarrow \\ \frac{5}{3} &= x \end{aligned}$$

16 Vul dit kruisgetallenraadsel in door de vergelijkingen op te lossen. In elk vakje noteer je één cijfer.

	A	B	C	D	E
A	4	1		1	1
B		3	1	2	
C	1		7	5	
D	2	9	0		2
E		6	0	3	8

HORIZONTAAL

A $4a-7=157$ • $3a-5=2a+6$

B $6x-500=4\cdot(x+31)$

C $2\cdot(x+2)-7=-x$ • $-3x=-x-150$

D $0,5x-15=2\cdot(0,2x+7)$ • $x+2-2x=0$

E $4\cdot(x-10\,000)=-3\cdot(x-700)+166$

VERTICAAL

A $-4\cdot(2x-5)=-3x$ • $\frac{1}{3}x+\frac{1}{6}x=9-\frac{1}{4}x$

B $19-2x=x-20$ • $\frac{1}{4}x+8=\frac{1}{3}x$

C $\frac{3}{5}\cdot\left(\frac{1}{20}x-45\right)-24=0$

D $0,5x-2,5=0,4x+10$ • $x-3=3-x$

E $4x+8-3x=5x+4$ • $\frac{1}{7}x+2=\frac{1}{2}x-\frac{1}{4}x-1$

a1
$$\begin{aligned} 4a-7 &= 157 \\ &\Updownarrow \\ 4a &= 164 \\ &\Updownarrow \\ a &= 41 \end{aligned}$$

a2
$$\begin{aligned} 3a-5 &= 2a+6 \\ &\Updownarrow \\ 3a-2a &= 11 \\ &\Updownarrow \\ a &= 11 \end{aligned}$$

b
$$\begin{aligned} 6x-500 &= 4(x+31) \\ &\Updownarrow \\ 6x-500 &= 4x+124 \\ &\Updownarrow \\ 6x-4x &= 124+500 \\ &\Updownarrow \\ 2x &= 624 \\ &\Updownarrow \\ x &= 312 \end{aligned}$$

c1
$$\begin{aligned} 2(x+2)-7 &= -x \\ &\Updownarrow \\ 2x+4-7 &= -x \\ &\Updownarrow \\ 2x+x &= 7-4 \\ &\Updownarrow \\ 3x &= 3 \\ &\Updownarrow \\ x &= 1 \end{aligned}$$

c2
$$\begin{aligned} -3x &= -x-150 \\ &\Updownarrow \\ -3x+x &= -150 \\ &\Updownarrow \\ -2x &= -150 \\ &\Updownarrow \\ x &= 75 \end{aligned}$$

d1
$$\begin{aligned} 0,5x-15 &= 2(0,2x+7) \\ &\Updownarrow \\ 0,5x-15 &= 0,4x+14 \\ &\Updownarrow \\ 0,5x-0,4x &= 15+14 \\ &\Updownarrow \\ 0,1x &= 29 \\ &\Updownarrow \\ x &= 290 \end{aligned}$$

d2
$$\begin{aligned} x+2-2x &= 0 \\ &\Updownarrow \\ -x &= -2 \\ &\Updownarrow \\ x &= 2 \end{aligned}$$

e
$$\begin{aligned} 4(x-10000) &= -3(x-700)+166 \\ &\Updownarrow \\ 4x-40000 &= -3x+2100+166 \\ &\Updownarrow \\ 4x+3x &= 40000+2100+166 \\ &\Updownarrow \\ 7x &= 42266 \\ &\Updownarrow \\ x &= 6038 \end{aligned}$$

a1
$$\begin{aligned} -4(2x-5) &= -3x \\ &\Updownarrow \\ -8x+20 &= -3x \\ &\Updownarrow \\ -8x+3x &= -20 \\ &\Updownarrow \\ -5x &= -20 \\ &\Updownarrow \\ x &= 4 \end{aligned}$$

a2
$$\begin{aligned} \frac{1}{3}x+\frac{1}{6}x &= 9-\frac{1}{4}x \\ &\Updownarrow \\ 4x+2x &= 108-3x \\ &\Updownarrow \\ 6x+3x &= 108 \\ &\Updownarrow \\ 9x &= 108 \\ &\Updownarrow \\ x &= 12 \end{aligned}$$

b1
$$\begin{aligned} 19-2x &= x-20 \\ &\Updownarrow \\ -2x-x &= -20-19 \\ &\Updownarrow \\ -3x &= -39 \\ &\Updownarrow \\ x &= 13 \end{aligned}$$

b2
$$\begin{aligned} \frac{1}{4}x+8 &= \frac{1}{3}x \\ &\Updownarrow \\ 3x+96 &= 4x \\ &\Updownarrow \\ 96 &= 4x-3x \\ &\Updownarrow \\ 96 &= x \end{aligned}$$

c
$$\begin{aligned} \frac{3}{5}\left(\frac{1}{20}x-45\right)-24 &= 0 \\ &\Updownarrow \\ \frac{3}{100}x-27-24 &= 0 \\ &\Updownarrow \\ 3x &= 2700+2400 \\ &\Updownarrow \\ 3x &= 5100 \\ &\Updownarrow \\ x &= 1700 \end{aligned}$$

d1
$$\begin{aligned} 0,5x-2,5 &= 0,4x+10 \\ &\Updownarrow \\ 5x-25 &= 4x+100 \\ &\Updownarrow \\ 5x-4x &= 100+25 \\ &\Updownarrow \\ x &= 125 \end{aligned}$$

d2
$$\begin{aligned} x-3 &= 3-x \\ &\Updownarrow \\ x+x &= 3+3 \\ &\Updownarrow \\ 2x &= 6 \\ &\Updownarrow \\ x &= 3 \end{aligned}$$

e1
$$\begin{aligned} 4x+8-3x &= 5x+4 \\ &\Updownarrow \\ x+8 &= 5x+4 \\ &\Updownarrow \\ 4 &= 4x \\ &\Updownarrow \\ x &= 1 \end{aligned}$$

e2
$$\begin{aligned} \frac{1}{7}x+2 &= \frac{1}{2}x-\frac{1}{4}x-1 \\ &\Updownarrow \\ 4x+56 &= 14x-7x-28 \\ &\Updownarrow \\ 4x+56 &= 7x-28 \\ &\Updownarrow \\ 4x-7x &= -28-56 \\ &\Updownarrow \\ -3x &= -84 \\ &\Updownarrow \\ x &= 28 \end{aligned}$$

17 Los de volgende vergelijkingen op in $\mathbb{Q}$ door de haakjes en noemers weg te werken.

a
$$6-\frac{x}{2} = \frac{x}{4}-4$$
$$\Updownarrow$$
$$\begin{aligned} 24-2x &= x-16 \\ &\Updownarrow \\ -2x-x &= -16-24 \\ &\Updownarrow \\ -3x &= -40 \\ &\Updownarrow \\ x &= \frac{40}{3} \end{aligned}$$

b
$$0,2x-\frac{2}{3}x = x-\frac{7}{15}$$
$$\Updownarrow$$
$$\begin{aligned} 3x-10x &= 15x-7 \\ &\Updownarrow \\ -7x &= 15x-7 \\ &\Updownarrow \\ 7 &= 22x \\ &\Updownarrow \\ x &= \frac{7}{22} \end{aligned}$$

c
$$\frac{1}{2}\cdot(x-5) = \frac{2}{5}\cdot\left(x-\frac{3}{4}\right)$$
$$\Updownarrow$$
$$\begin{aligned} \frac{1}{2}x-\frac{5}{2} &= \frac{2}{5}x-\frac{3}{10} \\ &\Updownarrow \\ 5x-25 &= 4x-3 \\ &\Updownarrow \\ 5x-4x &= -3+25 \\ &\Updownarrow \\ x &= 22 \end{aligned}$$

d
$$\frac{3}{4}x+4-\frac{1}{2}x = 5\cdot(3-x)$$
$$\Updownarrow$$
$$\begin{aligned} \frac{3}{4}x+4-\frac{1}{2}x &= 15-5x \\ &\Updownarrow \\ 3x+16-2x &= 60-20x \\ &\Updownarrow \\ x+16 &= 60-20x \\ &\Updownarrow \\ x+20x &= 60-16 \\ &\Updownarrow \\ 21x &= 44 \\ &\Updownarrow \\ x &= \frac{44}{21} \end{aligned}$$

e
$$\frac{1}{5}x-\frac{1}{10} = \frac{3}{2}\cdot\left(-x+\frac{33}{5}\right)$$
$$\Updownarrow$$
$$\begin{aligned} \frac{1}{5}x-\frac{1}{10} &= -\frac{3}{2}x+\frac{99}{10} \\ &\Updownarrow \\ 2x-1 &= -15x+99 \\ &\Updownarrow \\ 2x+15x &= 99+1 \\ &\Updownarrow \\ 17x &= 100 \\ &\Updownarrow \\ x &= \frac{100}{17} \end{aligned}$$

f
$$\frac{x}{2}+\frac{x}{3}-\frac{x}{4} = 6x-\frac{7}{12}$$
$$\Updownarrow$$
$$\begin{aligned} 6x+4x-3x &= 72x-7 \\ &\Updownarrow \\ 7x &= 72x-7 \\ &\Updownarrow \\ 7x-72x &= -7 \\ &\Updownarrow \\ -65x &= -7 \\ &\Updownarrow \\ x &= \frac{7}{65} \end{aligned}$$

18 Los de volgende vergelijkingen op in $\mathbb{Q}$.

a
$$3x-\frac{1}{2}\cdot\left(\frac{x}{2}+5\right) = 10+\frac{2}{3}x$$
$$\Updownarrow$$
$$\begin{aligned} 3x-\frac{x}{4}-\frac{5}{2} &= 10+\frac{2}{3}x \\ &\Updownarrow \\ 36x-3x-30 &= 120+8x \\ &\Updownarrow \\ 33x-8x &= 120+30 \\ &\Updownarrow \\ 25x &= 150 \\ &\Updownarrow \\ x &= 6 \end{aligned}$$

b
$$\frac{-3\cdot(x-2)}{5} = 4x-\frac{1}{5}$$
$$\Updownarrow$$
$$\begin{aligned} -3(x-2) &= 20x-1 \\ &\Updownarrow \\ -3x+6 &= 20x-1 \\ &\Updownarrow \\ 7 &= 23x \\ &\Updownarrow \\ x &= \frac{7}{23} \end{aligned}$$

c
$$3\cdot[2\cdot(3x-4)+5\cdot(x-5)] = -x-3$$
$$\Updownarrow$$
$$\begin{aligned} 3(6x-8+5x-25) &= -x-3 \\ &\Updownarrow \\ 18x-24+15x-75 &= -x-3 \\ &\Updownarrow \\ 33x-99 &= -x-3 \\ &\Updownarrow \\ 33x+x &= -3+99 \\ &\Updownarrow \\ 34x &= 96 \\ &\Updownarrow \\ x &= \frac{96}{34} \\ &\Updownarrow \\ x &= \frac{48}{17} \end{aligned}$$

d
$$\frac{2x-5}{3}-\frac{3x-2}{5} = \frac{3}{5}$$
$$\Updownarrow$$
$$\begin{aligned} 5(2x-5)-3(3x-2) &= 3\cdot 3 \\ &\Updownarrow \\ 10x-25-9x+6 &= 9 \\ &\Updownarrow \\ x-19 &= 9 \\ &\Updownarrow \\ x &= 28 \end{aligned}$$

e
$$-2\cdot(5x+3)+5\cdot(7-2x) = 0$$
$$\Updownarrow$$
$$\begin{aligned} -10x-6+35-10x &= 0 \\ &\Updownarrow \\ -20x &= 6-35 \\ &\Updownarrow \\ -20x &= -29 \\ &\Updownarrow \\ x &= \frac{29}{20} \end{aligned}$$

f
$$2-\frac{x-3}{2} = 1+\frac{x-3}{2}$$
$$\Updownarrow$$
$$\begin{aligned} 4-x+3 &= 2+x-3 \\ &\Updownarrow \\ 7-x &= x-1 \\ &\Updownarrow \\ 8 &= 2x \\ &\Updownarrow \\ x &= 4 \end{aligned}$$

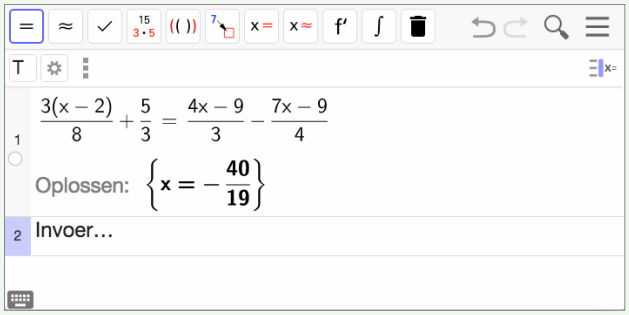
19 Om moeilijke vergelijkingen op te lossen, gebruik je het best ICT.

Voorbeeld: $\frac{3\cdot(x-2)}{8}+\frac{5}{3}=\frac{4x-9}{3}-\frac{7x-9}{4}$

Methode 1: stap voor stap oplossen met Photomath

Methode2: controle met Microsoft Math Solver

Methode 3: controle met de CAS van GeoGebra 6



Je merkt dat GeoGebra de oplossing noteert als een verzameling.

Los nu deze vergelijkingen op met ICT.

a $\frac{x}{6}-\frac{2x-1}{6}-\frac{1}{3}\cdot\left(\frac{2}{5}-\frac{x}{3}\right)=0$ $\qquad x=\frac{3}{5}$

b $\frac{3\cdot(2x+1)}{5}+\frac{x-1}{2}-1=\frac{5x}{3}$ $\qquad x=27$

c $\frac{5\cdot(2x-1)}{4}+\frac{3\cdot(x-5)}{2}=\frac{1}{4}$ $\qquad x=\frac{9}{4}$

d $\frac{1}{5}\cdot\left[\frac{1}{3}\cdot(10x-9)-\frac{5}{6}\cdot\left(\frac{2}{5}x-4\right)\right]=-\frac{7}{30}$ $\qquad x=-\frac{1}{2}$

20 Omgekeerd redeneren. Vervolledig de tabel om de onderstaande vergelijking op te lossen.

$$\frac{2\cdot(x+6)}{3}-4=2$$

WISKUNDETAAL	VERGELIJKING	REDENEREN
een getal x	$x = 3$	⟵ oplossing vergelijking
tel er 6 bij op	$x = 9-6$	trek er 6 van af
vermenigvuldig met 2	$x + 6 = 9$	deel door 2
deel door 3	$2\cdot(x+6) = 18$	vermenigvuldig met 3
trek er 4 van af	$\frac{2\cdot(x+6)}{3} = 6$	tel er 4 bij op
	$\frac{2\cdot(x+6)}{3}-4=2$	

21 We spreken af dat $a \heartsuit b$ betekent: $ab + a + b$. Bijvoorbeeld $5 \heartsuit 8 = 5\cdot 8 + 5 + 8 = 53$.
Er is een getal x waarvoor geldt: $3 \heartsuit 5 = 2 \heartsuit x$.

Welk getal is x?

(A) 3 (B) 6 (C) 7 (D) 10 (E) 12

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$3 \heartsuit 5 = 3\cdot 5 + 3 + 5 = 23$

$2 \heartsuit x = 2x + x + 2 = 3x + 2$

$$\begin{aligned} 3x+2 &= 23 \\ &\Updownarrow \\ 3x &= 21 \\ &\Updownarrow \\ x &= 7 \end{aligned}$$

22 Als ▲ + ▲ + 6 = ▲ + ▲ + ▲ + ▲, welk getal staat dan op de plaats van ▲?

(A) 2 (B) 3 (C) 4 (D) 5 (E) 6

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$$\begin{aligned} 2x+6 &= 4x \\ &\Updownarrow \\ 6 &= 2x \\ &\Updownarrow \\ 3 &= x \end{aligned}$$